

Previews

Financing a Green New Deal

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The Great Depression galvanized the New Deal in the United States to recover the economy, relieve the poor and unemployed, and reform the financial backbone of the country. Today, the impending adverse effects of climate change have governments proposing analogous Green New Deals. Recently in *Energy Research & Social Science*, Galvin and Healy critically examine the economic viability and wide-spanning societal, monetary, and environmental implications of the US Green New Deal proposals.

In the early 1930s, the world was in a severe economic recession—The Great Depression. Upon his 1932 Democratic presidential nomination, Franklin D. Roosevelt (commonly referred to by his nickname FDR) proposed a “new deal” to revitalize a battered United States economy. The term was first introduced by American humorist Mark Twain in his 1889 satire “A Connecticut Yankee in King Arthur’s Court.” It was here he purported that what his fictional exploited working class needed was a “new deal” aligned with their best interests. FDR’s campaign adapted the term to label his policies for recovering the economy, relieving the poor and jobless, and reforming the financial backbone of the country to avert future recessions.

The Great Depression occurred somewhat unexpectedly and suddenly, with economists still divided on its exact causes. Today, the world is experiencing another sudden, unexpected economic downturn because of the COVID-19 pandemic. Some problems cannot be predicted, and only broad, widely encompassing preventive measures can be set to mitigate damages. Other problems, however, have better-defined causes, outcomes, and timelines to avert them. One such example is the impending impacts of anthropogenic climate change.

Owing to rapid global industrialization from the late 1800s onward, the Earth is

warming. To prevent social and economic damages from increasing global mean temperatures, the Intergovernmental Panel on Climate Change (IPCC) estimates that warming must be limited to below 1.5°C of the pre-industrial era average.¹ The greenhouse gas byproducts of industrialized production that fed economic growth over the last century and a half are well ingrained into global economies today. To meet IPCC targets, the world must rapidly decarbonize all greenhouse gas emitting sectors and further, remove greenhouse gases from the air with negative emissions technologies.

Facing the imminent risks of climate change, members of the United States Democratic party are once again proposing to mobilize the country to overcome a threat. Congresswoman Alexandria Ocasio-Cortez and Senator Ed Markey proposed a Green New Deal that aims to decarbonize the country in an equitable and sustainable way. Senator Bernie Sanders put forth an adapted but fully costed Green New Deal as part of his 2020 United States presidential candidate platform. Recently in *Energy Research & Social Science*, Galvin and Healy offer a perspective on the economic viability and wide-spanning societal, monetary, and environmental implications of a Green New Deal.²

To understand the financial implications of a Green New Deal, Galvin and Healy

examine the proposal from the perspective of Keynesian or “demand-side” economics. Keynesian economics became prominent during and after the Great Depression and is named after British economist John Maynard Keynes. It describes the theory that it is the demand of goods and services, not supply, that drives economic output. This seemingly subtle distinction has an important impact on the implementation and justification of economic policy and the role that government has in its regulation. From the demand-side perspective, the pertinent question to ask when financing an initiative such as a Green New Deal is not “can we raise enough external funds to pay for it” (because the US is ultimately in control of its own sovereign currency) but rather, “will paying for this cause intolerable inflation?” They examine this question in more detail by considering Senator Sanders’s Green New Deal, as it is fully costed.

The Green New Deal proposed by Senator Sanders has a cost estimate of \$16.3 trillion (all subsequent dollar values herein are in USD) with a projected payback time of 15 years.³ These expenditures span 14 categories. Unsurprisingly, the majority of these funds—\$9,936 billion—are in the areas of renewable energy, energy efficiency, electric vehicles, and low-carbon research and development. After all, climate targets cannot be met without rapid decarbonization of energy production. The rest of the funds are allocated toward addressing infrastructure shortfalls and resiliency, transportation, social equality, sustainable food and water, and air quality. A complete cost breakdown is given in Figure 1A, with Figure 1B highlighting some of the specific energy expenditures across these areas.

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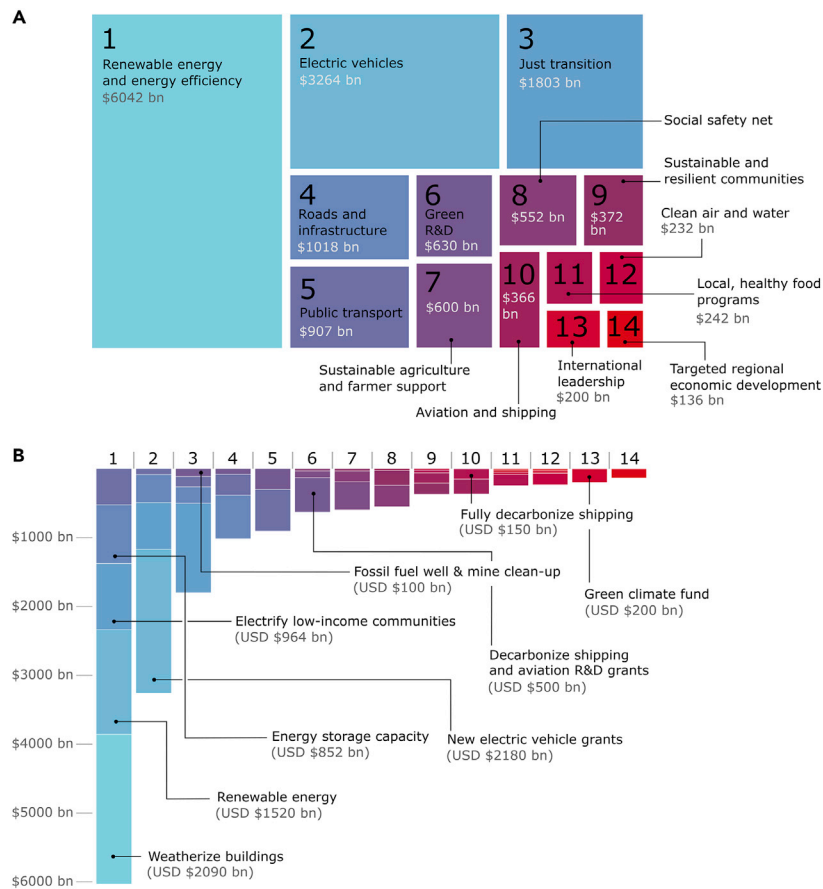


Figure 1. Cost Breakdown of Senator Bernie Sanders's Green New Deal

(A) The proposal spans 14 categories; their relative costs are shown here.

(B) Highlights of some of the specific energy costs associated with the proposal.

Payment of the Green New Deal comes from a restructuring of fossil fuel taxation, subsidies, and fines (\$3.085 trillion), reduced military expenditure in the oil sector (\$1.215 trillion), selling renewable electricity (\$6.4 trillion), income taxation on newly created jobs (\$2.3 trillion), public assistance program relief (\$1.31 trillion), and increased taxation of the wealthy and mega corporations (\$2 trillion). To understand the feasibility of this payment structure, Galvin and Healy first provide a critical examination of one of the most prevalent Keynesian economic critiques of the Green New Deal from Thomas Palley.⁴

Applying Palley's methodology but factoring in expected revenue from Senator Sanders's proposed Medicare-for-All (which is separate from the Green

New Deal but projected to save up to \$5.1 trillion over 10 years), and spanning a 15-year payback window, it was estimated that the total tax collected by the government would increase by 28.6%. This is lower than Sanders' own estimates of 40%, leading Galvin and Healy to conclude that Sanders's proposal is reasonable, if not an overestimate. So, would the country be willing to tolerate such an increase in tax take (which would predominantly impact high earners and corporations)? There is a historical precedent for similar taxation increases. During World War 2, government expenditure increased up to 45% of Gross Domestic Product, and average inflation during this period was around 5%. With proper management, expenditures like the Green New Deal would not necessarily lead to markedly increased inflation. It is

less a question of economic feasibility but rather one of social tolerance for mobilization to fight climate change.

At the heart of social motivation for climate change mitigation is the interplay between inequality, poverty, and carbon emissions. Galvin and Healy argue that individuals at economic extremes are responsible for a disproportionate amount of CO₂ emissions. Some studies have shown that the personal CO₂ emissions of the wealthiest 1% can be up to 20 times that of the US average.⁵ Poorer households in the European Union were found to also emit more than average due to unaffordability of energy efficiency upgrades.⁶ It is further argued that that a Green New Deal would inhibit top-emitting mega corporations from having lobbying power to keep their economic production (and thereby CO₂ emissions) maximized. Considering these points, pervasive economic inequality is not only a social issue, but an energy and climate issue as well. To offset potential cost burdens of a Green New Deal from falling on poorer households, over \$1,803 billion in Sanders's proposal is allocated toward a "just transition" to support the financial and medical wellbeing of individuals in high-risk sectors (such as fossil fuels), remediation of polluted sites, new job opportunities, and more. This further comes with \$552 billion in new social safety nets.

It was contended that the Green New Deal proposed by Senator Sanders could be financed without significant rises in inflation, provided that tax increases on top wealth brackets return to values seen throughout the 1940s to 1970s. An axiom of demand-side economics is that governments should increase expenditures during economic recessions. With the economic uncertainty surrounding the COVID-19 pandemic, it is perhaps an even more decisive time to explore large-scale mobilization efforts such as the Green New Deal. Further investigation of these elements in the energy social science research space will bring greater understanding toward a salient Green

New Deal that is just, socially equitable and economically feasible and also does its part to meet climate targets.

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Unlocking a Secret Stash of Energy

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In this issue of *Joule*, Dahn and coworkers present the development of a hybrid lithium-ion/lithium-metal cell that allows for an extra boost of capacity when needed. This battery has the potential to help eliminate range anxiety with electric vehicles.

When I was in college, lots of my friends had VW Bugs (Beetles), one of the most popular cars for young people in those days. That's because it was cheap (especially as a used car), everyone learned how to drive a stick shift on a Bug, and it even had remarkably good (for then) fuel economy, around 20 mpg. There was a drawback for the older models, however: the fuel gauge was so inaccurate that there was always the chance you'd run out of gas. But you didn't really have to worry, because the car had a "secret" reserve gallon that would get you to the gas station or get you home. I hadn't thought of that in many years, until I read this paper from Jeff Dahn's lab and started to wonder, what if electric cars had a secret stash of energy that provided an extra 50 miles of range?

The lithium battery world has long been focused on increasing energy density, but for some applications (cars, cell phones), gravimetric energy density—the focus of most academic work—is of secondary importance in comparison with volumetric energy density.

Consider a Tesla Model 3 with a 74-kWh battery pack made from Tesla/Panasonic 2170 cells with 246 Wh/kg and 711 Wh/l.¹ A 10% reduction in battery weight would save 30 kg, considerably less than 1% of the weight of the car. As such, the effect on range might well be too small to affect the value of the car. On the other hand, a 10% reduction in battery volume would save 10.5 L. This extra space could be used to enhance passenger comfort by providing additional knee and head room by enough to be noticed and appreciated. Furthermore, the volume reduction could make packaging the car mechanicals easier for the manufacturer, which, in turn, could save cost and weight.

The present work, which can be read in this issue of *Joule*, focuses on the volumetric energy density improvement possible from creating a hybrid Li-ion/Li-metal battery.² The improvement comes from borrowing an idea already in use but applied here in an interesting and novel way. In some current batteries, small amounts of Si are added to

graphite anodes to improve energy density. Although Si typically reduces the life of batteries, this effect can be mitigated when combined with graphite, because the delithiation potential of Si is higher than that of graphite. As a result, typical discharge cycles might not access the lithiated Si, reducing its effect on life while still providing extra energy when needed. The authors ask, can an analogous benefit be achieved by plating Li metal on graphite, providing extra energy—but only when needed? What impresses me about this work is that we are all taught from diapers that plating Li metal on graphite is a terrible thing. It is never easy to think outside the box and go against what we all "know," but these authors did.

The authors built cells with a substantial excess cathode capacity, reversing the usual situation, in which anode capacity is larger in order to reduce the possibility of plating. For this "hybrid cell," no plating occurred up to a charging voltage of 4.0 V, at which point the graphite was fully lithiated. Charging between 4.0 and 4.4 V accessed the remaining cathode Li, depositing it on the graphite anode as metallic Li. The result was a 20% to 25% improvement in volumetric energy density when charged to

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